



LEWIS PRESS  
LIMITED

Customer

**PHAROS**

End Customer

Generic Name

Product Reference No

Strength

Proof Date

Proof Version

Leaflet Dimensions

**PRS LFT100\_165x500**

Finish Folded Size

**165x250**

Leaflet Pharmacode



**1234**

Separation Colours

( FRONT )

( BACK )

**ONLY colours listed in this column will be printed**

PRF No

CIS Ref No

Proof Approval

DATE:

**INTRACETAM**  
Levetiracetam  
INTRACETAM 250 mg Film-coated Tablets  
INTRACETAM 500 mg Film-coated Tablets  
INTRACETAM 1000 mg Film-coated Tablets

**PHARMACEUTICAL FORM AND STRENGTH**

**Levetiracetam, 250 mg, film-coated tablets.**

Each film-coated tablet contains 250 mg of levetiracetam..  
Blue color, oval shaped film coated tablets with a break line on one side.

**Levetiracetam, 500 mg, film-coated tablets**

Each film-coated tablet contains 500 mg of levetiracetam..  
Yellow color, oval shaped film coated tablets with a break line on one side.

**Levetiracetam, 1000 mg, film-coated tablets**

Each film-coated tablet contains 1000 mg of levetiracetam .  
White color, oval shaped film coated tablets with a break line on one side.

**EXCIPIENTS**

**Levetiracetam, 250 mg, film-coated tablets**

Crospovidone (type B) (E1202), Povidone K30, Silica, colloidal anhydrous (E551), Magnesium stearate (E470b), Hypromellose (E464), Macrogol / PEG 400 (E1521), Titanium dioxide (E171), Talc (E553b), Indigo carmine lake (E132), Water, purified

**Levetiracetam, 500 mg, film-coated tablets**

Crospovidone (type B) (E1202), Povidone K30, Silica, colloidal anhydrous (E551), Magnesium stearate (E470b), Hypromellose (E464), Macrogol / PEG 400 (E1521), Titanium dioxide (E171), Talc (E553b), Iron oxide yellow (E172), Water, purified

**Levetiracetam, 1000 mg, film-coated tablets**

Crospovidone (type B) (E1202), Povidone K30, Silica, colloidal anhydrous (E551), Magnesium stearate (E470b), Hypromellose (E464), Macrogol / PEG 400 (E1521), Titanium dioxide (E171), Talc (E553b), Water, purified

**INDICATIONS**

Intracetam is indicated as monotherapy in the treatment of partial onset seizures with or without secondary generalisation in patients from 16 years of age with newly diagnosed epilepsy.

Intracetam is indicated as adjunctive therapy

- in the treatment of partial onset seizures with or without secondary generalisation in adults and children from 4 years of age with epilepsy.
  - in the treatment of myoclonic seizures in adults and adolescents from 12 years of age with Juvenile Myoclonic Epilepsy.
  - in the treatment of primary generalised tonic-clonic seizures in adults and children from 12 years of age with Idiopathic Generalised Epilepsy.
- Levetiracetam concentrate is an alternative for patients (adults and children from 4 years of age) when oral administration is temporarily not feasible.

**DOSAGE AND ADMINISTRATION**

Levetiracetam therapy can be initiated with either intravenous or oral administration. Conversion to or from oral to intravenous administration can be done directly without titration. The total daily dose and frequency of administration should be maintained.

**Film-coated tablets**

The film-coated tablets must be taken orally, swallowed with a sufficient quantity of liquid and may be taken with or without food. After oral administration the bitter taste of levetiracetam may be experienced. The daily dose is administered in two equally divided doses.  
The score line is only to facilitate breaking for ease of swallowing and not to divide into equal doses.

**Oral solution**

The oral solution may be diluted in a glass of water and may be taken with or without food. After oral administration the bitter taste of levetiracetam may be experienced. The daily dose is administered in two equally divided doses.

**Concentrate for solution for infusion**

Levetiracetam concentrate is for intravenous use only and the recommended dose must be diluted in at least 100 ml of a compatible diluent and administered intravenously as a 15- minute intravenous infusion (see Section Incompatibilities and Use and Handling).  
There is no experience with administration of intravenous levetiracetam for longer period than 4 days.  
Levetiracetam concentrate is an alternative for patients (adults and children from 4 years of age) when oral administration is temporarily not feasible.

**ROUTE OF ADMINISTRATION**

For oral use.

**ADULTS**

**MONOTHERAPY**

**Adults and adolescents from 16 years of age**  
The recommended starting dose is 250 mg twice daily which should be increased to an therapeutic dose of 500 mg twice daily after two weeks. The dose can be further increased by 250 mg twice daily every two weeks depending upon the clinical response. The maximum dose is 1500 mg twice daily.

**ADD-ON THERAPY**

**Adults (18 years) and adolescents (12 to 17 years) weighing 50 kg or more**  
The initial therapeutic dose is 500 mg twice daily. This dose can be started on the first day of treatment.  
Depending upon the clinical response and tolerability, the daily dose can be increased up to 1500 mg twice daily. Dose changes can be made in 500 mg twice daily increases or decreases every two to four weeks.

**CHILDREN**

The physician should prescribe the most appropriate pharmaceutical form, presentation and strength according to age, weight and dose.  
The tablet formulation is not adapted for use in children under the age of 6 years. Levetiracetam oral solution is the preferred formulation for use in this population. In addition, the available dose strengths of the tablets are not appropriate for initial treatment in children weighing less than 25 kg, for patients unable to swallow tablets or for the administration of doses below 250 mg. In all of the above cases levetiracetam oral solution should be used.  
The safety and efficacy of levetiracetam concentrate for solution for infusion in infants and children less than 4 years have not been established.

**MONOTHERAPY**

The safety and efficacy of levetiracetam in children and adolescents below 16 years as monotherapy treatment have not been established.

**ADD-ON THERAPY**

**Add-on therapy for children (4 to 11 years) and adolescents (12 to 17 years) weighing less than 50 kg**

Levetiracetam oral solution is the preferred formulation for use in children under the age of 6 years.  
For children 6 years and above, levetiracetam oral solution should be used for doses under 250 mg, for doses not multiple of 250 mg when dosing recommendation is not achievable by taking multiple tablets and for patients unable to swallow tablets.  
The initial therapeutic dose is 10 mg/kg twice daily.  
Depending upon the clinical response and tolerability, the dose can be increased up to 30 mg/kg twice daily. Dose changes should not exceed increases or decreases of 10 mg/kg twice daily every two weeks.  
The lowest effective dose should be used.  
Dose in children 50 kg or greater is the same as in adults.

**Dose recommendations for children and adolescents:**

| Weight                | Starting dose<br>10 mg/kg twice daily | Maximum dose<br>30 mg/kg twice daily |
|-----------------------|---------------------------------------|--------------------------------------|
| 10 kg <sup>(i)</sup>  | 100 mg (1 ml) twice daily             | 300 mg (3 ml) twice daily            |
| 15 kg <sup>(ii)</sup> | 150 mg (1.5 ml) twice daily           | 450 mg (4.5 ml) twice daily          |
| 20 kg <sup>(ii)</sup> | 200 mg (2 ml) twice daily             | 600 mg (6 ml) twice daily            |
| 25 kg <sup>(ii)</sup> | 250 mg twice daily                    | 750 mg twice daily                   |

(1) Children 25 kg or less should preferably start the treatment with levetiracetam 100mg/ml oral solution  
(2) Dose in children and adolescents 50 kg or more is the same as in adults.

**ELDERLY**

Adjustment of the dose is recommended in elderly patients with compromised renal function.

**RENAL IMPAIRMENT**

The daily dose must be individualised according to renal function (see Section Warnings and Precautions).  
For adult patients, refer to the following table and adjust the dose as indicated. To use this dosing table, an estimate of the patient's creatinine clearance (CLcr) in ml/min is needed. The CLcr in ml/min may be estimated from serum creatinine (mg/dl) determination, for adults and adolescents weighing 50 kg or more, using the following formula:

$$CLcr \text{ (ml/min)} = \frac{[140 - \text{age (years)}] \times \text{weight (kg)}}{72 \times \text{serum creatinine (mg/dl)}} \quad (\times 0.85 \text{ for women})$$

Then CLcr is adjusted for body surface area (BSA) as follows:

$$CLcr \text{ (ml/min/1.73m}^2\text{)} = \frac{CLcr \text{ (ml/min)}}{BSA_{\text{subject}} \text{ (m}^2\text{)}} \times 1.73$$

**Dosing adjustment for adult and adolescent patients weighing more than 50kg with impaired renal function:**

| Group                                                               | Creatinine clearance<br>(ml/min/1.73m <sup>2</sup> ) | Dosage and frequency                     |
|---------------------------------------------------------------------|------------------------------------------------------|------------------------------------------|
| Normal                                                              | ≥ 80                                                 | 500 to 1500 mg twice daily               |
| Mild                                                                | 50 - 79                                              | 500 to 1000 mg twice daily               |
| Moderate                                                            | 30 - 49                                              | 250 to 750 mg twice daily                |
| Severe                                                              | < 30                                                 | 250 to 500 mg twice daily                |
| End-stage renal disease patients undergoing dialysis <sup>(1)</sup> |                                                      | 500 to 1000 mg once daily <sup>(2)</sup> |

- (1) A 750 mg loading dose is recommended on the first day of treatment with levetiracetam.  
(2) Following dialysis, a 250 to 500 mg supplemental dose is recommended.

For children with renal impairment, levetiracetam dose needs to be adjusted based on the renal function as levetiracetam clearance is related to renal function.

This recommendation is based on a study in adult renally impaired patients. The CLcr in ml/min/1.73 m<sup>2</sup> may be estimated from serum creatinine (mg/dl) determination using, for young adolescents and children using the following formula (Schwartz formula):

$$CLcr \text{ (ml/min/1.73 m}^2\text{)} = \frac{\text{Height (cm)} \times \text{ks}}{\text{Serum Creatinine (mg/dl)}}$$

ks = 0.55 in Children to less than 13 years and in adolescent female; ks= 0.7 in adolescent male

**Dosing adjustment for children and adolescents patients weighing less than 50 kg withimpaired renal function:**

| Group                                                | Creatinine clearance<br>(ml/min/1.73m <sup>2</sup> ) | Children and adolescents<br>weighing less than 50 kg (1)      |
|------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------|
| Normal                                               | ≥ 80                                                 | 10 to 30 mg/kg (0.10 to 0.30 ml/kg) twice daily               |
| Mild                                                 | 50 - 79                                              | 10 to 20 mg/kg (0.10 to 0.20 ml/kg) twice daily               |
| Moderate                                             | 30 - 49                                              | 5 to 15 mg/kg (0.05 to 0.15 ml/kg) twice daily                |
| Severe                                               | < 30                                                 | 5 to 10 mg/kg (0.05 to 0.10 ml/kg) twice daily                |
| End-stage renal disease patients undergoing dialysis |                                                      | 10 to 20 mg/kg (0.10 to 0.20 ml/kg) once daily <sup>(2)</sup> |

- (1) Levetiracetam oral solution should be used for doses under 250 mg, for doses not multiple of 250 mg when dosing recommendation is not achievable by taking multiple tablets and for patients unable to swallow tablets.  
(2) A 15 mg/kg (0.15 ml/kg) loading dose is recommended on the first day of treatment with levetiracetam.  
(3) Following dialysis, a 5 to 10 mg/kg (0.05 to 0.10 ml/kg) supplemental dose is recommended.

**HEPATIC IMPAIRMENT**

No dose adjustment is needed in patients with mild to moderate hepatic impairment. In patients with severe hepatic impairment, the creatinine clearance may underestimate the renal insufficiency. Therefore a 50% reduction of the daily maintenance dose is recommended when the creatinine clearance is < 60 ml/min/1.73m<sup>2</sup>.

**CONTRAINDICATIONS**

Levetiracetam is contraindicated in:  
• Hypersensitivity to the active substance or other pyrrolidone derivatives or to any of the excipients.

**WARNINGS AND PRECAUTIONS**

**Discontinuation**

If levetiracetam has to be discontinued it is recommended to withdraw it gradually (e.g. in adults and adolescents weighing more than 50 kg; 500 mg decreases twice daily every two to four weeks; in children and adolescents weighing less than 50 kg: dose decrease should not exceed 10 mg/kg twice daily every two weeks).

**Renal or hepatic impairment**

The administration of levetiracetam to patients with renal impairment may require dose adjustment. In patients with severely impaired hepatic function, assessment of renal function is recommended before dose selection.

**Acute kidney injury**

The use of levetiracetam has been very rarely associated with acute kidney injury, with a time to onset ranging from a few days to several months.

**Blood cell counts**

Rare cases of decreased blood cell counts (neutropenia, agranulocytosis, leucopenia, thrombocytopenia and pancytopenia) have been described in association with levetiracetam administration, generally at the beginning of the treatment. Complete blood cell counts are advised in patients experiencing important weakness, pyrexia, recurrent infections or coagulation disorders (see Section Adverse Reactions).

**Depression and/or suicidal ideation**

Potential for an increase in risk of suicidal thoughts or behaviors.  
Suicide, suicide attempt, suicidal ideation and behaviour have been reported in patients treated with anti-epileptic agents (including levetiracetam).  
Therefore patients should be monitored for signs of depression and/or suicidal ideation and behaviours and appropriate treatment should be considered. Patients (and caregivers of patients) should be advised to seek medical advice if should signs of depression and/or suicidal ideation or behaviour emerge.

**Abnormal and aggressive behaviours**

Levetiracetam may cause psychotic symptoms and behavioural abnormalities including irritability and aggressiveness. Patients treated with levetiracetam should be monitored fordeveloping psychiatric signs suggesting important mood and/or personality changes. If such behaviours are noticed, treatment adaptation or gradual discontinuation should be considered. If discontinuation is considered, please see Section Discontinuation in Warnings and Precautions.

**Worsening of seizure**

As with other types of antiepileptic drugs, levetiracetam may rarely exacerbate seizure frequency or severity. This paradoxical effect was mostly reported within the first month after levetiracetam initiation or increase of the dose, and was reversible upon drug discontinuation or dose decrease. Patients should be advised to consult their physician immediately in case of aggravation of epilepsy.

**Electrocardiogram QT interval prolongation**

Rare cases of ECG QT interval prolongation have been observed during the post-marketing surveillance. Levetiracetam should be used with caution in patients with QTc-interval prolongation, in patients concomitantly treated with drugs affecting the QTc-interval, or in patients with relevant preexisting cardiac disease or electrolyte disturbances.

**Paediatric population**

The tablet formulation is not adapted for use in children under the age of 6 years. Available data in children did not suggest impact on growth and puberty. However, long term effects on learning, intelligence, growth, endocrine function, puberty and childbearing potential in children remain unknown.

**INTERACTION**

**Probenecid**

Probenecid (500 mg four times daily), a renal tubular secretion blocking agent, has been shown to inhibit the renal clearance of the primary metabolite, but not of levetiracetam. Nevertheless, the concentration of this metabolite remains low.

**Methotrexate**

Concomitant administration of levetiracetam and methotrexate has been reported to decrease methotrexate clearance, resulting in increased/prolonged blood methotrexate concentration to potentially toxic levels. Blood methotrexate and levetiracetam levels should be carefully monitored in patients treated concomitantly with the two drugs.

**Oral contraceptives, digoxin and warfarin**

Levetiracetam 1,000 mg daily did not influence the pharmacokinetics of oral contraceptives (ethinyl-estradiol and levonorgestrel); endocrine parameters (luteinizing hormone and progesterone) were not modified. Levetiracetam 2000 mg daily did not influence the pharmacokinetics of digoxin and warfarin; prothrombin times were not modified. Co-administration with digoxin, oral contraceptives and warfarin did not influence the pharmacokinetics of levetiracetam.

**Laxatives**

There have been isolated reports of decreased levetiracetam efficacy when the osmotic laxative macrogol has been concomitantly administered with oral levetiracetam. Therefore, macrogol should not be taken orally for one hour before and for one hour after taking levetiracetam.

**Food and alcohol**

The extent of absorption of levetiracetam was not altered by food, but the rate of absorption was slightly reduced.  
No data on the interaction of levetiracetam with alcohol are available.

**PREGNANCY AND LACTATION**

**Fertility**

No clinical data are available, potential risk for human is unknown.

**Women of childbearing potential**

Specialist advice should be given to women who are of childbearing potential. Treatment with levetiracetam should be reviewed when a woman is planning to become pregnant. As with all antiepileptic medicines, sudden discontinuation of levetiracetam should be avoided as this may lead to breakthrough seizures that could have serious consequences for the woman and the unborn child. Monotherapy should be preferred whenever possible because therapy with multiple antiepileptic medicines AEDs could be associated with a higher risk of congenital malformations than monotherapy, depending on the associated antiepileptics.

**Pregnancy**

Levetiracetam can be used during pregnancy, if after careful assessment it is considered clinically needed. In such case, the lowest effective dose is recommended. Physiological changes during pregnancy may affect levetiracetam concentration. Decrease in levetiracetam plasma concentrations has been observed during pregnancy. This decrease is more pronounced during the third trimester (up to 60% of baseline concentration before pregnancy). Appropriate clinical management of pregnant women treated with levetiracetam should be ensured.

**Lactation**

Levetiracetam is excreted in human breast milk. Therefore, breast-feeding is not recommended. However, if levetiracetam treatment is needed during breastfeeding, the benefit/risk of the treatment should be weighed considering the importance of breastfeeding.

**ABILITY TO PERFORM TASKS THAT REQUIRE JUDGEMENT, MOTOR OR COGNITIVE SKILLS**

Levetiracetam has minor or moderate influence on the ability to drive and use machines. Due to possible different individual sensitivity, some patients might experience somnolence or other central nervous system related symptoms, especially at the beginning of treatment or following a dose increase. Therefore, caution is recommended in those patients when performing skilled tasks, e.g. driving vehicles or operating machinery. Patients are advised not to drive or use machines until it is established that their ability to perform such activities is not affected.

**ADVERSE REACTIONS**

The most frequently reported adverse reactions were nasopharyngitis, somnolence, headache, fatigue and dizziness. The safety profile of levetiracetam is generally similar across age groups (adult and paediatric patients) and across the approved epilepsy indications. Adverse drug reactions (ADRs) are listed below by MedDRA system organ class and by frequency.

**Frequencies are defined as:**

- Very common Common Uncommon Rare
- Very rare
- Not known (cannot be estimated from the available data)

**Infections and infestations**

**Very common:** nasopharyngitis  
**Rare:** infection

**Blood and lymphatic system disorders**

**Uncommon:** thrombocytopenia, leukopenia  
**Rare:** neutropenia, pancytopenia, agranulocytosis

**Immune system disorders**

**Rare:** drug reaction with eosinophilia and systemic symptoms (DRESS); hypersensitivity (including angioedema and anaphylaxis)

**Metabolism and nutrition disorders**

**Common:** anorexia  
**Uncommon:** weight decreased, weight increase  
**Rare:** hyponatraemia

**Psychiatric disorders**

**Common:** depression, hostility/aggression, anxiety, insomnia, nervousness/irritability  
**Uncommon:** suicide attempt, suicidal ideation, psychotic disorder, abnormal behaviour, hallucination, anger, confusional state, panic attack, affect lability/mood swings, agitation  
**Rare:** completed suicide, personality disorder, thinking abnormal

**Nervous system disorders**

**Very common:** somnolence, headache  
**Common:** convulsion, balance disorder, dizziness, lethargy, tremor  
**Uncommon:** amnesia, memory impairment, coordination abnormal/ataxia, paraesthesia, disturbance in attention  
**Rare:** choreoathetosis, dyskinesia, hyperkinesia, gait disturbance

**Eye disorders**

**Uncommon:** diplopia, vision blurred

**Ear and labyrinth disorders**

**Common:** vertigo

**Cardiac disorders**

**Rare:** electrocardiogram QT prolonged

**Respiratory, thoracic and mediastinal disorders**

**Common:** cough

**Gastrointestinal disorders**

**Common:** abdominal pain, diarrhoea, dyspepsia, nausea, vomiting  
**Rare:** pancreatitis

**Hepatobiliary disorders**

**Uncommon:** liver function test abnormal  
**Rare:** hepatic failure, hepatitis

**Renal and urinary disorders**

**Rare:** acute kidney injury

**Skin and subcutaneous tissue disorders**

**Common:** rash  
**Uncommon:** alopecia, eczema, pruritus  
**Rare:** toxic epidermal necrolysis, Stevens-Johnson syndrome, erythema multiforme

**Musculoskeletal and connective tissue disorders**

**Uncommon:** muscular weakness, myalgia  
**Rare:** rhabdomyolysis and blood creatine phosphokinase increased\*

**General disorders and administration site conditions**

**Common:** asthenia/fatigue

**Injury, poisoning and procedural complications**

**Uncommon:** injury

\* Prevalence is significantly higher in Japanese patients when compared to non-Japanese patients.

**Description of selected adverse reactions**  
The risk of anorexia is higher when levetiracetam is coadministered with topiramate. In several cases of alopecia, recovery was observed when levetiracetam was discontinued. Bone marrow suppression was identified in some of the cases of pancytopenia. Cases of encephalopathy have been rarely observed after levetiracetam administration. These undesirable effects generally occurred at the beginning of the treatment (few days to a few months) and were reversible after treatment discontinuation.

**Paediatric population**

The adverse event profile of levetiracetam is generally similar across age groups and across the approved epilepsy indications. Safety results in paediatric patients were consistent with the safety profile of levetiracetam in adults except for behavioural and psychiatric adverse reactions which were more common in children than in adults. In children and adolescents aged 4 to 16 years, vomiting (very common), agitation (common), mood swings (common), affect lability (common), aggression (common); abnormal behaviour (common) and lethargy (common) were reported more frequently than in other age ranges or in the overall safety profile. In infants and children aged 1 month to less than 4 years, irritability (very common) and coordination abnormal (common) were reported more frequently than in other age groups or in the overall safety profile.

**OVERDOSAGE**

**Symptoms and signs**

Somnolence, agitation, aggression, depressed level of consciousness, respiratory depression and coma were observed with levetiracetam overdoses.

**Treatment**

There is no specific antidote for levetiracetam. Treatment of an overdose will be symptomatic and may include haemodialysis. The dialyser extraction efficiency is 60 % for levetiracetam and 74 % for the primary metabolite. Further management should be as clinically indicated or as recommended by the national poisons centre, where available.

**CLINICAL PHARMACOLOGY**

**Pharmacodynamics**

**Pharmacotherapeutic group**  
Antiepileptics; Other Antiepileptics

**ATC Code**  
N03AX14

**Mechanism of Action**

The active substance, levetiracetam, is a pyrrolidone derivative (S-enantiomer of  $\alpha$ -ethyl- 2-oxo-1-pyrrolidine acetamide), chemically unrelated to existing antiepileptic active substances. The mechanism of action of levetiracetam still remains to be fully elucidated. Levetiracetam does not alter basic cell characteristics and normal neurotransmission. In vitro studies show that levetiracetam affects intraneuronal Ca<sup>2+</sup> levels by partial inhibition of N-type Ca<sup>2+</sup> currents and by reducing the release of Ca<sup>2+</sup> from intraneuronal stores. In addition it partially reverses the reductions in GABA- and glycine-gated currents induced by zinc and  $\beta$ -carbolines. Furthermore, levetiracetam has been shown in in vitro studies to bind to a specific site in rodent brain tissue. This binding site is the synaptic vesicle protein 2A, believed to be involved in vesicle fusion and neurotransmitter exocytosis. Levetiracetam and related analogs show a rank order of affinity for binding to the synaptic vesicle protein 2A which correlates with the potency of their anti-seizure protection in the mouse audiogenic model of epilepsy. This finding suggests that the interaction between levetiracetam and the synaptic vesicle protein 2A seems to contribute to the antiepileptic mechanism of action of the medicinal product.

**Pharmacodynamic effects**

Levetiracetam induces seizure protection in a broad range of animal models of partial and primary generalised seizures without having a pro-convulsant effect. The primary metabolite is inactive. In man, an activity in both partial and generalised epilepsy conditions (epileptiform discharge/photoparoxysmal response) has confirmed the broad spectrum pharmacological profile of levetiracetam.

**Pharmacokinetics**

Levetiracetam is a highly soluble and permeable compound. The pharmacokinetic profile is linear with low intra- and intersubject variability. There is no modification of the clearance after repeated administration. The time independent pharmacokinetic profile of levetiracetam was also confirmed following 1500 mg intravenous infusion for 4 days with twice daily dosing. There is no evidence for any relevant gender, race or circadian variability. The pharmacokinetic profile is comparable in healthy volunteers and in patients with epilepsy.

Due to its complete and linear absorption, plasma levels can be predicted from the oral dose of levetiracetam expressed as mg/kg bodyweight. Therefore there is no need for plasma level monitoring of levetiracetam.

A significant correlation between saliva and plasma concentrations has been shown in adults and children (ratio of saliva/plasma concentrations ranged from 1 to 1.7 for oral tablet formulation and after 4 hours post-dose for oral solution formulation).

The pharmacokinetic profile has been characterized following oral administration. A single dose of 1500 mg levetiracetam diluted in 100 ml of a compatible diluent and infused intravenously over 15 minutes is bioequivalent to 1500 mg levetiracetam oral intake, given as three 500 mg tablets.

The intravenous administration of doses up to 4000 mg diluted in 100 ml of 0.9% sodium chloride infused over 15 minutes and doses up to 2500 mg diluted in 100 ml of 0.9% sodium chloride infused over 5 minutes was evaluated. The pharmacokinetic and safety profiles did not identify any safety concerns.

**Absorption**

Levetiracetam is rapidly absorbed after oral administration. Oral absolute bioavailability is close to 100%. Peak plasma concentrations (C<sub>max</sub>) are achieved at 1.3 hours after dosing. Steady-state is achieved after two days of a twice daily administration schedule. Peak concentrations (C<sub>max</sub>) are typically 31 and 43 µg/ml following a single 1,000 mg dose and repeated 1,000 mg twice daily dose, respectively. The extent of absorption is dose- independent and is not altered by food.

**Distribution**

No tissue distribution data are available in humans. Neither levetiracetam nor its primary metabolite are significantly bound to plasma proteins (<10 %). The volume of distribution of levetiracetam is approximately 0.5 to 0.7 l/kg, a value close to the total body water volume.

**Metabolism**

Levetiracetam is not extensively metabolised in humans. The major metabolic pathway (24 % of the dose) is an enzymatic hydrolysis of the acetamide group. Production of the primary metabolite, ucb L057, is not supported by liver cytochrome P450 isoforms. Hydrolysis of the acetamide group was measurable in a large number of tissues including blood cells. The metabolite ucb L057 is pharmacologically inactive. Two minor metabolites were also identified. One was obtained by hydroxylation of the pyrrolidone ring (1.6 % of the dose) and the other one by opening of the pyrrolidone ring (0.9 % of the dose). Other unidentified components, accounted only for 0.6 % of the dose.

No enantiomeric interconversion was evidenced in vivo for either levetiracetam or its primary metabolite.

In vitro, levetiracetam and its primary metabolite have been shown not to inhibit the major human liver cytochrome P450 isoforms (CYP3A4, 2A6, 2C9, 2C19, 2D6, 2E1 and 1A2), glucuronyl transferase (UGT1A1 and UGT1A6) and epoxide hydroxylase activities. In addition, levetiracetam does not affect the in vitro glucuronidation of valproic acid.

In human hepatocytes in culture, levetiracetam had little or no effect on CYP1A2, SULT1E1 or UGT1A1. Levetiracetam caused mild induction of CYP2B6 and CYP3A4. The in vitro data and in vivo interaction data on oral contraceptives, digoxin and warfarin indicate that no significant enzyme induction is expected in vivo. Therefore, the interaction of levetiracetam with other substances, or vice versa, is unlikely.

**Elimination**

The plasma half-life in adults was 7±1 hours and did not vary either with dose, route of administration or repeated administration. The mean total body clearance was 0.96 ml/min/kg.

The major route of excretion was via urine, accounting for a mean 95 % of the dose (approximately 93 % of the dose was excreted within 48 hours. Excretion via faeces accounted for only 0.3 % of the dose.

The cumulative urinary excretion of levetiracetam and its primary metabolite accounted for 66 % and 24 % of the dose, respectively during the first 48 hours. The renal clearance of levetiracetam and ucb L057 is 0.6 and 4.2 ml/min/kg respectively indicating that eyetiracetam is excreted by glomerular filtration with subsequent tubular reabsorption and that the primary metabolite is also excreted by active tubular secretion in addition to glomerular filtration. Levetiracetam elimination is correlated to creatinine clearance.

**Special patient populations**

**Children**

Children (4 to 12 years)  
Following single oral dose administration (20 mg/kg) to epileptic children (6 to 12 years), the half-life of levetiracetam was 6.0 hours. The apparent body weight adjusted clearance was approximately 30% higher than in epileptic adults. Following repeated oral dose administration (20 to 60 mg/kg/day) to epileptic children (4 to 12 years), levetiracetam was rapidly absorbed. Peak plasma concentration was observed 0.5 to 1.0 hour after dosing. Linear and dose proportional increases were observed for peak plasma concentrations and area under the curve. The elimination half-life was approximately 5 hours. The apparent body clearance was 1.1 ml/min/kg.

**Elderly**

In the elderly, the half-life is increased by about 40 % (10 to 11 hours). This is related to the decrease in renal function in this population.

**Renal impairment**

The apparent body clearance of both levetiracetam and of its primary metabolite is correlated to the creatinine clearance. It is therefore recommended to adjust the maintenance daily dose of levetiracetam, based on creatinine clearance in patients with moderate and severe renal impairment.

In anuric end-stage renal disease subjects the half-life was approximately 25 and 3.1 hours during interdialytic and intradialytic periods, respectively.

The fractional removal of levetiracetam was 51 % during a typical 4-hour dialysis session.

**Hepatic impairment**

In subjects with mild and moderate hepatic impairment, there was no relevant modification of the clearance of levetiracetam.

In most subjects with severe hepatic impairment, the clearance of levetiracetam was reduced by more than 50% due to a concomitant renal impairment.

**STORAGE CONDITION AND EXPIRY DATE**

3 years. Store below 30°C.

Keep out of reach and sight of children.

Do not use after the expiry date stated on the carton box.

**NATURE AND CONTENTS OF CONTAINER**

Intracetam 250mg, 500mg and 1000mg film-coated tablets are packaged in PVC-PE- PVDC/aluminium blisters placed into cardboard boxes containing 60 film-coated tablets.

**MANUFACTURER**

Larriss Hellas Medical and Pharmaceutical Products S.A.  
LARRISA INDUSTRIAL AREA, P.O. BOX 3012,  
LARRISA, 41 500, GREECE

Current at Dec 2024